

JUNOH JUNG

Postdoctoral Appointee, Mathematics and Computer Science Division, Argonne National Laboratory

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ACADEMIC APPOINTMENTS AND RESEARCH EXPERIENCE

Postdoctoral Appointee, MCS Division, Argonne National Laboratory Advisor: Emil Constantinescu. Mentor: Bethany Lusch. Research topic: scientific machine learning for PDE-based simulations (CFD) using high-performance computing.	2024.Nov–present
Graduate Student Research Assistant, Mechanical Engineering, University of Michigan Advisor: Aaron Towne. Research topic: resolvent-based estimation and control of aerodynamic flows.	2019.Aug–2024.Aug
Givens Associate, MCS Division, Argonne National Laboratory Supervisor: David Lenz. PI: Thomas Peterka. Research topic: physics-informed B-spline framework for flow data.	2023
Master Thesis Student, Aerodynamisches Institut Aachen (AIA), RWTH Aachen University Advisor: Wolfgang Schroeder. Research topic: development of a fully coupled zonal RANS/LES method for simulation of a turbulent backward-facing step flow.	2017–2018
Visiting Scholar, Department of Aerospace Engineering, Georgia Institute of Technology Supervisor: Dimitri Mavris.	2017

EDUCATION

Ph.D., Mechanical Engineering, University of Michigan Dissertation: Resolvent-based Estimation and Control of Aerodynamic Flows. Advisor: Aaron Towne.	2019-2024
M.S., Aerospace Engineering, RWTH Aachen University Thesis: Development of a fully coupled zonal RANS/LES method for the simulation of a turbulent backward-facing step flow. Advisor: Wolfgang Schroeder.	2014-2018
B.S., Mechanical and System Design Engineering, Hongik University Cumulative rank: 1/166 (highest graduating rank in the department).	2008-2014

HONORS, FELLOWSHIPS, AND AWARDS

Argonne Leadership Computing Facility (ALCF) Postdoctoral Fellowship, Argonne National Laboratory	Nov 2024
Argonne Training Program for Extreme-Scale Computing (ATPESC), funded by the U.S. Department of Energy	Aug 2023
Rackham Travel Grant, University of Michigan	Winter 2022; Winter 2023
Mechanical Engineering Fellowship, University of Michigan	Winter 2020
Academic excellence scholarships (full tuition and second highest), Hongik University	2009-2014

PUBLICATIONS

Peer-reviewed journal articles

- Jung, J.**, Balin, R., Constantinescu, E. Agent-orchestrated multi-fidelity workflow for a stable hybrid physics-machine learning simulation. In preparation.
- Jung, J.**, Balin, R., Lusch, B., Constantinescu, E., Accelerating a Spectral Element Solver for Large-Scale Computing with a Hybrid Physics–Machine Learning Framework. To be submitted.
- Jung, J.**, Constantinescu, E. Weak-form correction to accelerate finite element solvers. To be submitted.
- Jung, J.**, Lenz, D., Constantinescu, E., Peterka, T. (2026). Physics-Informed B-spline Reconstruction of Flow Data. *Computer Methods in Applied Mechanics and Engineering*. Under review.
- Jung, J.**, Towne, A. (2026). Resolvent-based estimation of turbulent wakes. *Journal of Fluid Mechanics*.
- Jung, J.**, Bhagwat, R., Towne, A. (2025). Resolvent-based estimation and control of laminar flow over an airfoil. *Journal of Fluid Mechanics*.
- Farghadan, A., **Jung, J.**, Bhagwat, R., Towne, A. (2024). Efficient harmonic resolvent analysis via time-stepping. *Theoretical and Computational Fluid Dynamics*.

8. Martini, E., **Jung, J.**, Cavalieri, A. V. G., Jordan, P., Towne, A. (2022). Resolvent-based tools for optimal estimation and control via the Wiener-Hopf formalism. *Journal of Fluid Mechanics*.

Conference proceedings

1. **Jung, J.**, Constantinescu, E., Balin, R. (2027). Agent-Orchestrated Multi-Fidelity Workflow for Stable Hybrid Physics–Machine Learning Simulations. *AIAA SciTech Forum*. (accepted)
2. **Jung, J.**, Constantinescu, E., Balin, R., Lusch, B. (2026). A hybrid physics–machine-learning framework for enhancing a coarse-grid spectral element solver for large-scale computing. *AIAA Aviation Forum*.
3. **Jung, J.**, Constantinescu, E. (2026). Learning Differentiable Weak-Form Corrections to Accelerate Finite Element Simulations. *ASME FEDSM*.
4. Zhou, Y., Towne, A., **Jung, J.**, Bhagwat, R., Martini, E., Jordan, P., Audiffred, D., Maia, I., Cavalieri, A. V. G. (2026). Towards resolvent-based estimation and control of wavepackets in supersonic turbulent jets. *AIAA SciTech Forum*.
5. Towne, A., Bhagwat, R., Zhou, Y., **Jung, J.**, Martini, E., Jordan, P., Audiffred, D. B. S., Maia, I., Cavalieri, A. V. G. (2024). Resolvent-based estimation of wavepackets in turbulent jets. *AIAA/CEAS 2024*.
6. Towne, A., **Jung, J.**, Bhagwat, R., Zhou, Y., Paulson, J. (2024). Resolvent-based estimation and control of wakes and jets. *13th International Symposium on Turbulence and Shear Flow Phenomena*.
7. **Jung, J.**, Towne, A. (2024). Toward turbulent wake estimation using a resolvent-based approach. *AIAA SciTech Forum*.
8. **Jung, J.**, Bhagwat, R., Towne, A. (2023). Resolvent-based estimation of laminar flow around an airfoil. *AIAA SciTech Forum*.

INVITED TALKS

1. **Jung, J.** (2027). From weak-form correction to Agent-orchestrated workflow for hybrid physics-ML framework, SIAM CSE27. Mini symposium titled "Recent Advances in Galerkin Methods for Physical Systems"
2. **Jung, J.** (2024). Estimation and Control of Aerodynamic Flows Using a Resolvent-Based Approach, ALCF.

SELECTED TALKS AND PRESENTATIONS

1. **Jung, J.**, Constantinescu, E., Balin, R. (2027). Agent-Orchestrated Multi-Fidelity Workflow for Stable Hybrid Physics–Machine Learning Simulations. *AIAA SciTech Forum*. (expected)
2. **Jung, J.**, Constantinescu, E. (2026). Towards agentic workflows for automated and stable hybrid physics–ML simulations. *APS Division of Fluid Dynamics, oral presentation*.
3. **Jung, J.**, Constantinescu, E. (2026). A hybrid physics–machine-learning framework for enhancing a coarse-grid spectral element solver for large-scale computing. *Nek User Meeting*.
4. **Jung, J.**, Constantinescu, E., Balin, R., Lusch, B. (2026). A hybrid physics–machine-learning framework for enhancing a coarse-grid spectral element solver for large-scale computing. *AIAA Aviation*.
5. **Jung, J.**, Constantinescu, E. (2026). Learning Differentiable Weak-Form Corrections to Accelerate Finite Element Simulations. *ASME FEDSM*. Oral presentation
6. **Jung, J.**, Constantinescu, E. (2026). Differentiable Weak-Form Learning for Coarse-Mesh Incompressible Flow Simulation: Toward Chemical Engineering Applications, *18th AIChE Midwest regional conference*, oral presentation
7. **Jung, J.**, Constantinescu, E. (2026). Differentiable Weak Form Learning for Structure Preserving Finite Element Acceleration of Incompressible Flows. *USNCTAM26*, oral presentation.
8. **Jung, J.** (2026). Toward machine-learning-augmented resolvent-based estimation. *KSEA MRC 26*, poster presentation.
9. **Jung, J.**, Constantinescu, E. (2025). Hybrid Physics-Machine Learning Framework Toward Efficient Simulation of Turbulent Flows on an Exascale Platform. *APS Division of Fluid Dynamics*, oral presentation.
10. **Jung, J.**, Constantinescu, E. (2025). Hybrid Physics-Machine Learning Modeling with post-update discrete corrections. *Postdoctoral Research and Career Symposium*, Argonne National Laboratory, poster presentation.
11. **Jung, J.**, Towne, A. (2024). Resolvent-based estimation and control of airfoil wakes. *APS Division of Fluid Dynamics*, oral presentation.
12. **Jung, J.**, Towne, A. (2024). Toward turbulent wake estimation using a resolvent-based approach. *AIAA SciTech Forum*.
13. **Jung, J.**, Towne, A. (2023). Resolvent-based estimation for turbulent wakes. *APS Division of Fluid Dynamics*, oral presentation.
14. **Jung, J.** (2023). Resolvent-based estimation and control of flows, *Michigan Fluid Mechanics Research Seminar (internal)*, oral presentation.
15. **Jung, J.**, Bhagwat, R., Towne, A. (2023). Resolvent-based estimation of laminar flow around an airfoil. *AIAA SciTech Forum*.
16. **Jung, J.**, Bhagwat, R., Towne, A. (2022). Real-time estimation and control of flow over a NACA0012 airfoil using resolvent-based tools. *APS Division of Fluid Dynamics*, oral presentation.
17. **Jung, J.** (2021). Resolvent-based flow estimation and control, *Michigan Fluid Mechanics Research Seminar (internal)*, oral presentation.

18. Jung, J., Martini, E., Cavalieri, A., Jordan, P., Lesshafft, L., Towne, A. (2020). Optimal resolvent-based estimation for flow control. *APS Division of Fluid Dynamics*, oral presentation.

TALKS AS CONTRIBUTORS

1. Zhou, Y., Bhagwat, R., Jung, J., Towne, A. (2026). Resolvent-based Estimation and Control of Wavepackets in a Supersonic Jet. *APS Division of Fluid Dynamics*, oral presentation.
2. Zhou, Y., Jung, J., Bhagwat, R., Towne, A. (2025). Estimation and closed-loop control of supersonic jet noise. *APS Division of Fluid Dynamics*, oral presentation.
3. Constantinescu, E., Kang, S., Jung, J. (2025). Hybrid machine learning for multiscale PDE simulation using neural ODEs. *18th U.S. National Congress on Computational Mechanics*, oral presentation.
4. Towne, A., Jung, J., Bhagwat, R., Zhou, Y., Paulson, J. (2024). Resolvent-based estimation and control of wakes and jets. *13th International Symposium on Turbulence and Shear Flow Phenomena*, oral presentation.
5. Bhagwat, R., Jung, J., Martini, E., Schmidt, O., Jordan, P., Towne, A. (2022). Toward resolvent-based estimation and control of high-speed turbulent jets. *APS Division of Fluid Dynamics*, oral presentation.
6. Martini, E., Jung, J., Cavalieri, A. V. G., Jordan, P., Towne, A., Lesshafft, L. (2021). Optimal control for large systems: a resolvent-based approach. *XXV ICTAM Conference*, oral presentation.

GRANTS & FUNDING

- **Genesis Mission proposal led by Arizona State University (Phase 1, not selected)** Apr 2026
Project: AI-Enabled Design of Electrolyte Wetting Processes in Thick Porous Battery Electrodes
Role: Key Personnel
Focus: AI-enabled scientific workflows and adaptive learning methods for energy applications.
- **Laboratory Directed Research and Development, Argonne National Laboratory** Apr-Sep 2026
Project: Agent-Orchestrated Multi-Fidelity Workflow for Stable Hybrid Physics–Machine Learning Simulations
Role: Lead Principal Investigator (Sole PI)
Amount: \$25k
Focus: AI-for-science workflows, agentic adaptation, hybrid physics–ML simulation, and multi-fidelity scientific computing.
- **Director’s Discretionary Allocation Program, Argonne Leadership Computing Facility** 2026
Project: Differentiable Hybrid Physics Machine Learning Simulations
Role: Lead Principal Investigator (Sole PI)
Total node-hours: 45,000 hours, 6 months
Focus: Leadership-computing workflows for scalable scientific ML, model adaptation, and PDE-based simulation.

TEACHING EXPERIENCE

- Graduate Student Instructor, Introduction to Fluid Mechanics (ME 320), University of Michigan** 2024
Held office hours, conducted recitations, reviewed quizzes, and graded exams for 101 students for two lectures (ME320-001 and ME320-002)

FORMAL MENTOR

- GPU HACKATHONS at Argonne (member and mentor) 2025-2026

PROFESSIONAL SERVICE AND MEMBERSHIPS

Member, American Institute of Chemical Engineers (AIChE)	2026
Member, American Society of Mechanical Engineers (ASME)	2026
Member, American Institute of Aeronautics and Astronautics (AIAA)	2022-2026
Member, American Physical Society (APS)	2020-2026
Proposal reviewer, APEX program at Argonne	2026
Reviewer, Journal of Fluid Mechanics	2026
Reviewer, Scientific Reports	2026
Abstract Reviewer, AIAA SciTech (Fluid Dynamics: Flow Control; Applied Aerodynamics)	2024-2026

TECHNICAL EXPERTISE

- Programming and development: C/C++, Python, Fortran, MPI programming and parallel implementation (OpenMPI, MPICH), Git/GitHub, Ubuntu and bash.
- Scientific computing and modeling: CFD simulation (DNS, LES, RANS), reduced-order and data-driven modeling (resolvent-based approaches, DMD, POD), artificial intelligence and deep learning.
- Libraries and environments: PETSc, FFTW, LAPACK; GPU and heterogeneous computing; DOE, DoD, and university supercomputing environments.
- Visualization and design: Tecplot, ParaView, VisIt, UG-NX, AutoCAD.

LEADERSHIP AND ADDITIONAL PROFESSIONAL ACTIVITIES

Hyundai Motors Vision Conference, invited participant, Invited to South Korea for a talk with UAM engineers and factory visits.	Aug 2022
Local President, Korean-German Scientists and Engineers Association Elected local president for the Aachen, Cologne, and Bonn region.	Oct 2014-Oct 2016
Republic of Korea Army, Sergeant Role: machine gunner	Feb 2010-Dec 2011